HMGT 327 · FUNDAMENTALS OF PROJECT MANAGEMENT

Project Budgeting & Cost Estimation

Assignment 3 · Summary Sheet · Academic Year 1448 H (2026–2027 G), Fall 2026

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1 Why Project Budgeting Matters

01 Planning
Turns scope & schedule into money

02 Control baseline
Compare actual vs. plan; catch overruns early

03 Funding & cash flow
Money ready when contractors must be paid

04 Resource priorities
Spend first on guest-facing areas

05 Decisions
Approve, compare options, judge ROI

06 Risk
Contingency & management reserves

07 Accountability
One reference for owner, operator, contractors

Hospitality
Hajj-season room renovations, new restaurants, resort pre-opening

2 Financial Estimating Methods

Analogous (top-down)
Cost of a similar past project. Fast, less accurate.

Parametric
Cost per key / m² / seat

`120 rooms × SAR 60K = 7.2M`

Bottom-up
Estimate each WBS work package, then add up. Most accurate.

Three-point (PERT)
Reduces uncertainty

`(O + 4M + P) ÷ 6`

Expert judgment
Quantity surveyors, designers, technical services

Vendor bid analysis
Compare supplier quotes, e.g., FF&E

Reserve analysis
Contingency (known risks) + management reserve

Best practice
Top-down for approval, bottom-up for the detailed budget

Accuracy improves as scope is defined: **Rough order of magnitude −25% / +75%** → **Definitive −5% / +10%**

3 Steps & Techniques for Managing the Budget (PMI, 2017)

1 · Plan
Rules, units, accuracy, thresholds

2 · Estimate
WBS-based cost of every activity

3 · Budget
Add up + contingency = cost baseline (S-curve)

4 · Fund
Match spending to owner's funding & cash flow

5 · Control
Track actuals, forecast, change control

Earned Value Management (EVM)

CV = EV − AC · **SV** = EV − PV
CPI = EV ÷ AC · **SPI** = EV ÷ PV (< 1.0 = problem)
EAC = BAC ÷ CPI · plus variance & trend analysis, TCPI

Example: restaurant renovation, BAC SAR 2.0M

Month 3: PV 900K · EV 800K · AC 1.0M
CV **−200K** · SV **−100K** · CPI **0.80** · SPI **0.89**
EAC = 2.0M ÷ 0.80 = **SAR 2.5M** → act now

4 Types of Project Budgets

- **Top-down vs. bottom-up:** set by management vs. built from work estimates
- **Capital (CAPEX) vs. operating (OPEX):** building works, FF&E vs. salaries, utilities
- **Resource budgets:** labour, materials, OS&E, technology
- **Pre-opening:** recruitment, training, marketing, launch
- **Time-phased / cash-flow · fixed vs. flexible · reserves**

5 Guidelines for an Effective Budget

- ✓ **Clear scope & WBS** first
- ✓ **Involve** the people who do the work, finance & operator
- ✓ **Use historical data** and cross-check methods
- ✓ **Include all costs:** indirect, permits, fees, 15% VAT, inflation
- ✓ **Risk-based reserves** & documented assumptions
- ✓ **Time-phase, approve & baseline**; control changes
- ✓ **Monitor, report, update**; record lessons learned

Key sources: PMI (2017, 2021) · Kerzner (2022) · Larson & Gray (2021) · Meredith, Shafer & Mantel (2017) · Fleming & Koppelman (2010)